KPN Fiber Availability Analysis

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1 Address Data Collection and Sampling

I used the publicly available PDOK Locatieserver API to collect dutch addresses which provides address information and geographic coordinates. I selected nine geographically distributed cities: Utrecht, Amsterdam, The Hague, Eindhoven, Groningen, Rotterdam, Arnhem, Zwolle and Leeuwarden. For each city, I retrieved up to 100 address records and randomly selected 10 addresses which resulted in 90 different addresses.

Before sampling, I validated Dutch postal codes, checked for missing address information and removed duplicate addresses based on postcode, house number, and optional additions. I also extracted latitude and longitude to support the coordinates of geographic visualization.

The final address dataset was saved as a CSV file containing seven fields: postcode, house number, addition, street, city, latitude, and longitude.

2 KPN Fiber Status Retrieval and Classification

To check the fiber status on each adress, I used Python and Playwright to automate address lookups on the KPN Glasvezelnetwerk website. For each address, the script enters the postcode, house number and addition into the website's form and submits the request.

After receiving the response, I extracted the address displayed by KPN and compared it with the original address from PDOK. To account for formatting differences, I normalized punctuation, hyphens, and common street name abbreviations before comparing the addresses.

Since KPN provides fiber availability information through descriptive Dutch messages, I developed a rule-based classification system to convert these responses into five standardized categories. This approach makes the results easier to analyze, compare, and visualize geographically.

An address was classified as **Active** when the response indicated that fiber was already available, using phrases such as "jouw woning heeft glasvezel van KPN". The **Planned** category was assigned when future installation was indicated, such as "jouw woning krijgt glasvezel" or "we maken plannen voor de aanleg van glasvezel". The **Currently Installing** category was assigned when KPN indicated ongoing installation work, such as "legt glasvezel aan in jouw buurt". Addresses were classified as **Not Available** when fiber could not currently be provided, for example, "geen glasvezel beschikbaar". Finally, **Unknown** was assigned when none of the predefined phrases matched the response.

Each address was checked once, with a three-second delay between requests. Unsuccessful lookups and address mismatches were recorded separately to avoid assigning incorrect fiber availability statuses. The results were saved incrementally in a CSV file after each lookup.

3 Findings, Limitations, and Data Reliability

The geographic visualization revealed differences in KPN fiber availability across the selected cities. One of the most interesting finding was Rotterdam, where I expected a higher proportion of active fiber connections due to its size and urban development. However, the sampled addresses showed more planned connections than expected. In comparison, Eindhoven and Arnhem appeared to have a higher proportion of active fiber connections.

I ran the data collection and checked the fiber process five times. Consequently it helped assess whether the observed patterns were consistent across different samples and identify potential issues in the retrieval process. However, the findings have several limitations. The addresses were randomly selected from only the first 100 records returned by PDOK for each city, which introduced sampling bias. The sample is therefore not representative of either the selected cities or the Netherlands as a whole. Consequently, the findings should be interpreted as exploratory observations rather than definitive conclusions about fiber coverage in the Netherlands.

4 Scalability and Extension to Other Providers

The main challenge in scaling the solution to hundreds of thousands of addresses is retrieving fiber availability from KPN rather than collecting addresses from PDOK. The current implementation processes addresses sequentially which requires a separate website interaction for each lookup.

To improve efficiency, I would prioritize using an official KPN API or structured dataset, if available. Additionally, caching previously retrieved results would prevent unnecessary repeated lookups. Processing addresses in batches could improve performance and reliability. A database such as PostgreSQL could replace CSV files to efficiently store, manage and update large datasets.

To extend the solution to other fiber providers, I would create a shared system for managing addresses, storing results and handling errors. Since each provider (KPN, Odido) may provide information through different websites or APIs, the retrieval process would be adapted to each provider. All results would then be converted into standardized availability categories.

5 Time Spent and AI Tool Usage

I spent a total of 8 hours completing the case study. During development, I used AI models to assist with troubleshooting Playwright browser automation and improving code readability and documentation.

However, I independently reviewed and adapted the AI generated suggestions, tested the application's functionality and repeated the data collection and availability checks five times to examine the results and identify potential inconsistencies.